

Business Document Ingestion

Client Overview

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Purpose

Business Document Ingestion turns a collection of business records into a traceable, reviewable data package. It is designed for invoices, purchase orders, statements, commission reports, receipts, remittances, shipping documents, credit memos, and related records.

The system does more than read text. It preserves the original evidence, records how a value was found, checks that values are internally consistent, and makes uncertainty visible. The goal is to give your team organized data that can be traced back to the original page before it is used for reporting, reconciliation, or a later system load.

Courtesy credit is given to Christopher Melhauser and `theonlymuffinbot`. `theonlymuffinbot` describes collaborative AI tooling and is not a claim of AI legal authorship or ownership.

What the system is designed to deliver

At the end of a successful run, your team receives a package that can include:

- preserved source documents and page-level fingerprints;
- extracted fields, line items, tables, and document classifications;
- source citations and raw provider outputs for every AI-assisted proposal;
- arithmetic, validation, attribution, completeness, and final-review results;
- an Excel workbook, CSV exports, HTML reviewer views, and canonical JSON;
- a clear exception list for anything that is missing, ambiguous, or not yet authorized; and
- an approved-record staging or retrieval plan when the required gates are clear.

The system does not silently turn uncertainty into a passing result. A missing reference, conflicting reading, or unavailable accounting control remains an explicit exception until it is resolved or formally retained as an exception.

The central rule: evidence first

Every original page and extracted value is retained. A later correction is an append-only amendment linked to its evidence; it does not overwrite history. This approach allows a reviewer to answer four practical questions for any important value:

1. Where did this value come from?
2. Which page or source row supports it?
3. Which checks did it pass or fail?
4. What changed, who authorized it, and when?

AI, mappings, groupings, inferred relationships, and client workbook imports are proposals until an authorized operator applies an append-only change and the affected controls are rerun.

How the workflow works

The process is staged so a later phase cannot erase an earlier failure.

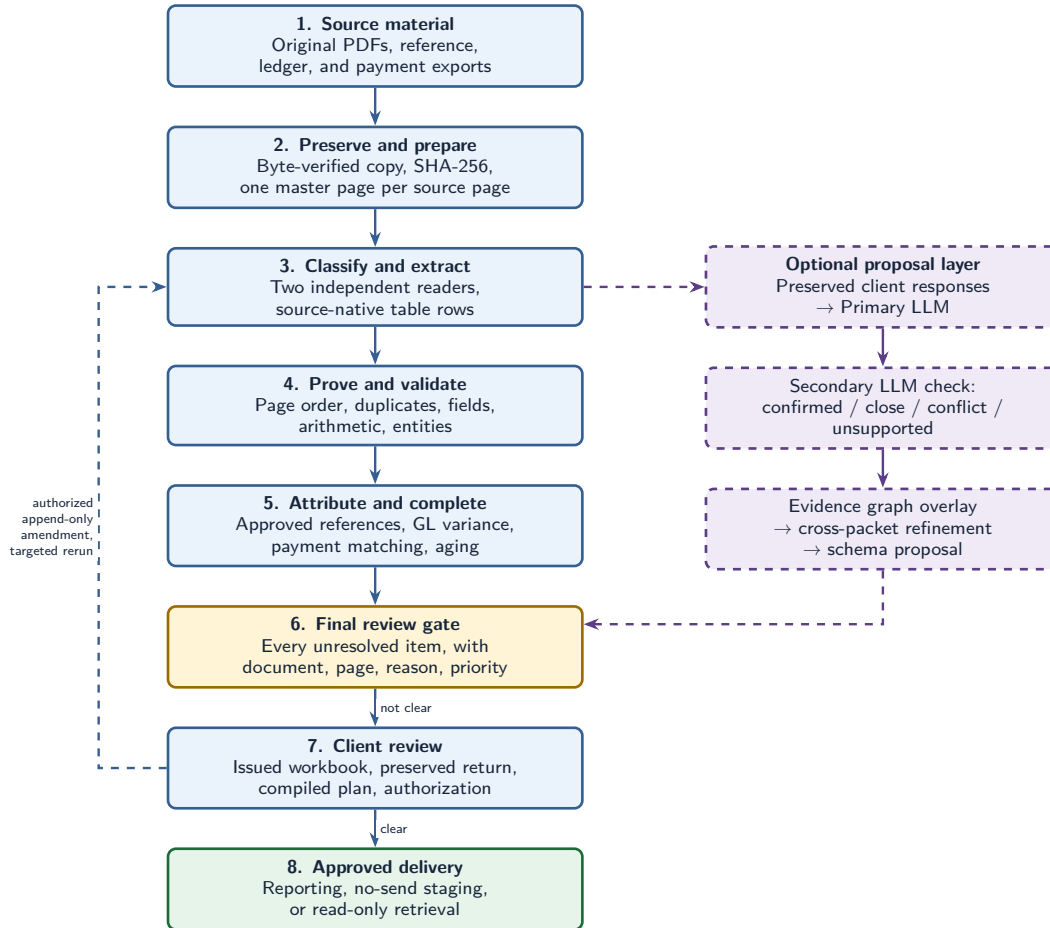


Figure 1: Client lifecycle from original evidence to an approved handoff. The deterministic spine (blue) is the only path that can clear a control; the optional proposal layer (purple, dashed) can only add reviewable proposals.

The eight stages below match the eight boxes in the diagram above. Each one states what the stage does, how it does it, and what it produces, so this section can be read on its own.

Step 1: The source material you supply

What it does. Establishes the evidence base. Everything the system can later prove is bounded by what arrives at this step.

How it does it. You supply the original source PDFs, unedited, in the order you received them, together with any available supporting exports: approved ACK, job, project, PO or order references; general-ledger data for the matching periods; and payment or remittance exports. Exports are read with the column headings your system already prints, so renaming columns is unnecessary and actively harmful — it edits the evidence before it is read.

What it produces. The delivery itself becomes retained evidence. Where an export does not exist, that absence is recorded as a known limitation rather than being worked around silently.

Step 2: Preserve and prepare

What it does. Freezes exactly what was received so that any later finding can be traced to a specific delivered page.

How it does it. The system keeps the original source, creates a SHA-256 fingerprint, verifies the retained copy byte-for-byte, and records a manifest of what was received. PDFs are separated into ordered one-page masters. For difficult scans, the workflow keeps the grayscale master and creates enhanced or black-and-white variants beside it; a profiling pass records contrast, skew, and compression concerns and sets the handwriting branch.

What it produces. A verified source copy, one immutable master per page, an intake manifest binding every page to its source file and position, comparison image variants, and a scan profile. Image improvements are aids for reading; they never replace the original source page.

Step 3: Classify and extract

What it does. Determines what each document is and reads the values printed on it.

How it does it. The system identifies document families and extracts printed fields, headers, totals, dates, parties, identifiers, and source-native table rows — invoice and acknowledgement numbers, purchase orders, jobs, projects, payment references, commission rates, customer names, and amounts. Where two genuinely independent extraction providers are configured, each reads the same page separately and the readings are compared; the system refuses two lanes that resolve to a single model vendor, because that is not independent evidence. Tabular documents are additionally read by a source-native table pass that follows the layout the document actually uses.

What it produces. Extraction proposals linked to their exact page, a consensus result recording agreement and disagreement, and retained raw provider responses. Agreement is useful evidence; disagreement, a single reading, or a provider failure remains review work. A provider's confidence score alone is not proof.

Step 4: Prove and validate

What it does. Tests whether the documents hold together internally, before anything is compared against outside systems.

How it does it. The workflow checks page order and grouping, surfaces possible duplicates, validates required fields and formats, and proves arithmetic against source-visible values — line extensions, taxes, discounts, freight, credits, subtotals, and document totals. It resolves entities and addresses through a reversible merge log, and checks readings for plausibility without inferring a unit, date order, or country from magnitude.

What it produces. An arithmetic proof result, a validation result, a reassembly proposal, and a duplicate-candidate list — each recording what was checked and what could not be proved. If a number does not add up, a page sequence cannot be proved, or a handwritten change conflicts with printed information, the issue stays visible. The system does not repair it by guessing, and a control that had nothing to check is reported as failed rather than clear.

Step 5: Attribute and test completeness

What it does. Connects amounts to your business references, then measures what the documents alone cannot establish.

How it does it. When your business process requires an amount to be associated with an ACK, job, project, purchase order, order, claim, or other reference, the system applies the reference hierarchy you

approve. Where general-ledger and payment or remittance exports are available, it compares document totals by accounting period, matches invoices to payments including partial payments, and computes aging.

What it produces. An attribution result with a full evidence chain per amount; an unattributable register listing every amount that could not be linked, with its reason and value; and a completeness report with period variances, payment matches, and aging status. Unmatched rows are reported back with their original heading and content rather than dropped. Unmatched amounts are never assigned to a convenient but unsupported reference, and a variance is reported rather than plugged.

Authoritative ledger and payment records are categorically different from document evidence, and are required to close the corresponding controls. Where they are absent, see the inferred-proposal section below.

Step 6: The final review gate

What it does. Collects everything unresolved into one queue and decides whether the run may advance.

How it does it. Every review-bearing artifact from every preceding stage — including recovery artifacts — is gathered into a single exhaustive queue. Items may be grouped by root cause so one decision can answer many similar cases, but grouping never removes an underlying item. Financially material issues are prioritized.

What it produces. The canonical `final_client_review.json` queue plus CSV, Excel, and HTML views of it, and a final reconciliation manifest stating which controls passed, which were not applicable, and which remain blocked. This manifest — not the appearance of the spreadsheet — is the authoritative answer to whether the run can advance.

Step 7: Client review and authorization

What it does. Captures your decisions without letting any of them apply themselves.

How it does it. The client-review package identifies the records that need a decision. When a workbook returns, the issued and returned copies are preserved separately and unchanged. The system validates the workbook structure against the issued copy, records every response and note into a preserved-response snapshot, and verifies that no response was omitted or altered. An operator then compiles a change plan stating exactly which mapping, amendment, or policy change is affected, which documents and fields are involved, and which downstream controls must rerun. Each proposed change is authorized individually, bound to the unchanged plan; anything not chosen stays deferred rather than being treated as a silent no.

What it produces. A preserved-response record, a compiled change plan, an operator authorization bound to that plan, and — only then — applied append-only amendments. This keeps client feedback visible without allowing a returned workbook to bypass the control process.

Step 8: Rerun and approved delivery

What it does. Re-proves everything downstream of an authorized change, then produces the handoff.

How it does it. Approved amendments are append-only: the original value remains in the audit trail beside the amendment. The affected extraction or mapping lane is rerun, followed by arithmetic, validation, attribution, completeness, and final review, in that order. Reruns are written to a new output location so an earlier manifest or package is never silently replaced.

What it produces. A regenerated package showing both the original client response and the resulting control status, and — only when the gate is clear or the remaining items are explicitly retained as approved exceptions — a canonical export, load plan, vendor-neutral no-send staging package, or read-only retrieval store.

What each stage produces, at a glance

Stage	Principal outputs	Can it clear a control?
1. Source material	Retained delivery; recorded gaps	No — it defines what is provable
2. Preserve and prepare	Verified copy, page masters, intake manifest, scan profile	No
3. Classify and extract	Extraction proposals, consensus result, raw responses	No — proposals only
4. Prove and validate	Arithmetic proof, validation result, duplicate and grouping proposals	Yes, for arithmetic and validation
5. Attribute and complete	Attribution result, unattributable register, completeness report	Yes, where authoritative exports exist
6. Final review gate	Canonical review queue, reviewer views, reconciliation manifest	It reports the gate
7. Client review	Preserved responses, compiled plan, authorization, amendments	No — authorization is not clearance
8. Rerun and delivery	Regenerated package, canonical export, staging or retrieval	Yes, on rerun of the affected controls

How the optional AI-assisted relationship process works

The system can use AI to find relationships that deterministic extraction may have missed. This is useful when no more client data will arrive and the source documents contain printed ACK, job, project, PO, invoice, order, payment, remittance, party, or template clues.

The AI process is bounded and evidence-based:

1. The system constructs a packet from retained source records and preserved client responses. Client comments are reasoning context, not source proof.
2. The Primary LLM proposes only source-visible relationships and must cite the supporting document and evidence quote.
3. An independently configured Secondary LLM checks the same packet and labels each proposal **confirmed**, **close_needs_review**, **conflict**, or **unsupported**.
4. Confirmed proposals are added as immutable graph overlays; they do not change source evidence or canonical data.
5. Later passes examine unresolved evidence neighborhoods. The process stops when a pass produces no material new, source-backed, Secondary-LLM-confirmed relationship or reaches its configured limit.
6. A graph-guided cross-packet pass examines related documents that could not be matched within a single packet. Its configured neighborhood has explicit document and byte limits, with separate exceptions for deferred or identifier-less records.

This process can improve reference discovery, relationship modeling, and understanding of repeated layouts. It cannot create an authoritative ledger balance, payment-system transaction, or client authorization from document evidence alone.

Schema discovery: how the system learns the structure

After retained extraction and relationship evidence are available, the system can build a proposal-only schema inventory. It identifies observed field names, counts, examples, document/template families, repeated line structures, and possible relationships.

The schema-discovery process can propose that a printed label likely represents a PO, ACK, customer, commission amount, or other business concept. It does not silently establish that mapping. A high-confidence proposal still needs clear source-template evidence, independent Secondary LLM review where configured, and the normal authorization path before it becomes a reusable rule.

What happens when client ledger, payment, or reference exports are missing

The system can build clearly labeled inferred-control proposals from retained document evidence. These may include source-linked attribution candidates, document-total rollups, explicit printed payment markers, and a detailed exception register.

These artifacts prioritize work and explain missing evidence. They do not substitute for a general ledger, payment system, or approved reference export; the final manifest keeps the corresponding control blocked unless authoritative evidence or an explicitly accepted exception is available.

The output: what you receive and how to read it

The optional development extension can retain original PNG/JPEG page images and let a connected vision-capable assistant propose schema-mapped fields. Receipts, unknown labels, uncertainties, corrections and rejections remain visible. A valid submission means **pending review, not published**; it cannot update the approved database or a CRM. The operator can export a private source archive and, for complete usable sources, a hash-linked image PDF for normal profiling/intake. Independent verification, business controls and authorization still apply. There is no bundled mobile capture app or attachment relay, and actual upload/vision support needs client acceptance. Governed record writes and expanded custom analytics remain development work. The output guide explains these capabilities in context; Visual Intake defines the technical limits.

This section is the summary of the deliverable. It describes what is in the package, what each part is for, how to read a result correctly, and what the package deliberately declines to claim.

For a self-contained explanation of the approved-data package, reports, MCP/API access, controlled downloads, and CRM-ready inputs, see `CLIENT_OUTPUT_OVERVIEW.md`. That guide is the plain-language bridge from this delivery package to safe day-to-day use.

What is in the package

- 1. The canonical review record.** `final_client_review.json` is the complete machine-readable queue and the authoritative list of what needs a decision. Each item identifies the document and page, the field, region or relationship in question, the reason it is open, its priority, and the evidence supporting the proposed reading.
- 2. The reviewer views.** `final_client_review.csv`, `.xlsx`, and `.html` present that same queue in forms that are easier to filter, sort, and discuss. They are convenience copies, not separate records; where they disagree with the JSON, the JSON governs. The Excel workbook carries **Header Definitions**, **Schema Definitions**, and **Runbook** sheets so it can be understood without another file.
- 3. The final reconciliation manifest.** This reports which controls passed, which were not applicable, and which remain blocked. It is the authoritative answer to whether the run can advance.
- 4. The evidence and audit trail.** Retained source documents and page-level fingerprints; extracted fields, line items, tables, and classifications; source citations and raw provider outputs for every AI-assisted proposal; the unattributable register and exception registers; preserved client responses; and every authorized amendment with its operator and timestamp.
- 5. The onward package, when the gate permits it.** A canonical export, load plan, vendor-neutral no-send CRM/API staging package, or citation-backed read-only retrieval store.

How to read a result correctly

- A passing control means the control ran and proved something.** A control that processed nothing is reported as failed, not clear, so a green result is never an artifact of an empty input.
- A blocked control names what is missing.** If no ledger or payment export was supplied, completeness stays blocked and says so. An inferred proposal cannot substitute for it, and the status `blocked_inferred_controls_not_authoritative` is not renamed or removed.

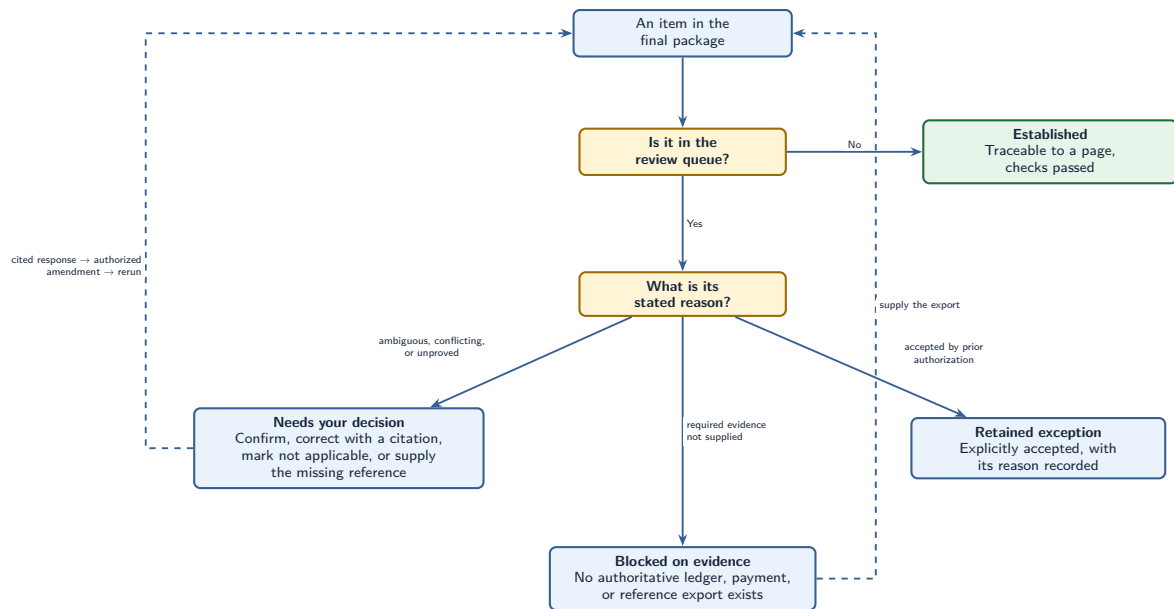


Figure 2: How to read any item in the final package. The reason attached to an item, not its confidence score, determines what is required to resolve it.

- **An exception is a result, not a process failure.** The system is designed to end with a precise list of what remains unresolved rather than a confident answer it cannot support.
- **A confidence score is provenance, never a decision term.** Neither a high score nor agreement between two readers performs the authorization step.
- **Every value is traceable.** For any value you can ask where it came from, which page supports it, which checks it passed or failed, and what changed, who authorized it, and when.

For each open item, choose one outcome

- Confirm the displayed value or grouping.
- Provide the corrected value, with the supporting page or system reference.
- Mark the item as not applicable, with a reason.
- Provide the missing reference, payment, or ledger record.

A response should identify the supporting source or system record wherever possible. Corrections are recorded as amendments; the original page and the original reading remain available for audit.

What the package deliberately does not claim

The delivery will not tell you that a missing general ledger reconciled, that an unmatched amount belongs to a plausible-looking reference, that a handwritten financial change was accepted because two readers agreed, or that an AI proposal is approved. Each of those remains an open item with its reason attached.

A clear review queue also does not prove that a required source, ledger, payment, or authorization step was supplied in the first place, and it is exhaustive only for the source artifacts listed inside that package. The final package states both what has been established **and** what remains unavailable, unresolved, or awaiting approval. Those two statements together are the deliverable.

What your team needs to provide

The best input is the original source PDF set plus any available supporting exports. Helpful inputs include:

- approved ACK, job, project, PO, order, or other business-reference exports;

- general-ledger data for the same periods, when reconciliation is required;
- payment or remittance exports, when invoice payment status or aging matters;
- a list of the business questions the resulting data must answer; and
- any original or higher-quality scans for pages that are difficult to read.

If one of these sources does not exist, tell the operator. The workflow can continue with explicit inference and exception controls, but it will not claim that a missing authoritative control has passed.

What requires your review

The final package focuses your attention on records that need a business decision. Common examples include:

- a document type, date, party, or page sequence that cannot be established;
- a conflicting provider or source reading;
- a handwritten correction or unclear printed value;
- a total that does not reconcile within the document;
- an amount with no approved business reference;
- a payment or ledger condition that cannot be verified; and
- a proposed template or schema mapping that needs approval.

The package prioritizes financially material issues while retaining every source item in the audit record.

What happens after you respond

Your issued workbook is preserved unchanged, and the returned workbook is stored separately. Before an operator interprets a response, the system copies every choice and note into a preserved-response record and checks that no response has been omitted or altered. The returned workbook itself remains evidence; it is never used to overwrite the original source document.

Next, the operator compiles an impact plan. It states exactly which proposed mapping, amendment, or policy change would be affected, which documents and fields are involved, and which downstream controls must run again. An authorized operator chooses each proposed change separately. Deferred or unsupported choices remain open rather than being treated as a silent no.

The affected stages are rerun in a new output directory. The system then rechecks downstream evidence, arithmetic, validation, attribution, completeness, and final review. The regenerated package shows both the original client response and the resulting control status, giving your team a clear before-and-after audit trail.

Privacy, access, and deployment boundaries

Source material remains under the run's evidence controls. Optional external services are disabled unless explicitly enabled for an approved run. Their results are proposals or corroboration, not automatic truth. Credentials stay outside the repository and are not written into a client delivery package.

The release can produce a vendor-neutral staging package and an approved-facts-only local retrieval service. Production CRM delivery, public hosting, or claimed accuracy rates require separate authorization and evidence.

When delivery is complete

A delivery is complete only when the named final-review package is clear and the applicable controls have passed or are explicitly retained as approved exceptions. Depending on the scope, this includes arithmetic, attribution, completeness, sampling, and final-review gates. The delivery identifies the exact source set, run manifest, approved amendments, exceptions, and any staging or retrieval plan so it can be independently reviewed later.

A clear review queue does not prove that a required source, ledger, payment, or authorization step was omitted. The final package states both what has been established and what remains unavailable, unresolved, or awaiting approval.